From Snapshots to Streams: A Live State Graph Approach for LLM-Based Root-Cause-Analysis in Kubernetes

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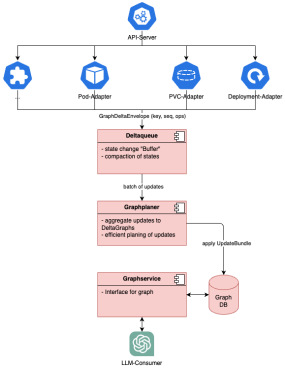

Outline

1. Introduction
2. Background
3. Research Question
4. Methodology
5. Evaluation
6. Contributions

Introduction

- Kubernetes, Containerisation and cloud-native technologies have become industry standard
- Key challenges for operation: perceived complexity, lack of training [3]

- IT-Downtime generate enormous costs for companies [8]
- Root cause analysis (RCA) as key bottleneck [5]

$\Rightarrow$ LLM-based assistance systems promise cognitive reduction and timely detection of issues

Background

Kubernetes (k8s):

- Open-source, feature-rich container orchestrator
- Deployments through declarative manifests
- Controller continuously reconcile actual with desried states

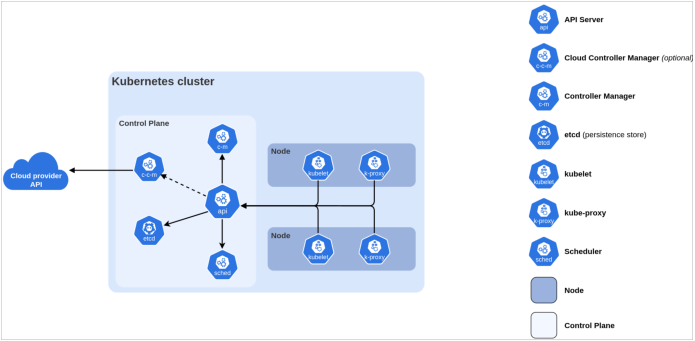

Figure 1: Kubernetes architecture

Background

Error sources:

- Unforeseen events: node dies, network issues, ...
- Syntactically correct but semantically incorrect manifests

Error handling:

- Reading docs
- Manually debugging through API-resources with the CLI
- Applying fixes

```yaml
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-config
data:
  key: value
---
apiVersion: v1
kind: Pod
metadata:
  name: my-app
  labels:
    app: my-app
spec:
  containers:
    - name: main
      image: nginx:1.27
      volumeMounts:
        - name: config
          mountPath: /etc/config
  volumes:
    - name: config
      configMap:
        name: app-config-typo
```

Figure 2: Incorrect manifest

Background

Synergy-RCA [9]:

- periodically maps cluster snapshots to graph
- leverages LLMs to generate queries, checks, reports

Limitations:

- graph is not live → doesn't capture all events
- heavylifting of LLM prompts and expert knowledge

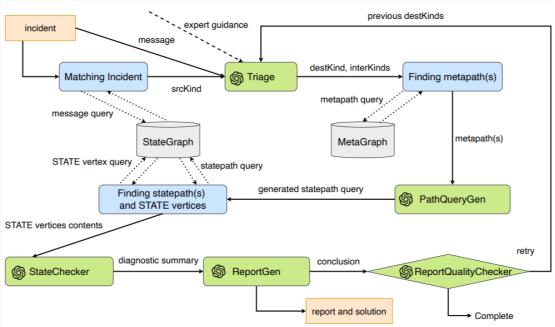

Figure 3: SynergyRCA Architecture

Research Question

Research Question:

"Does a live state graph for Kubernetes and predefined traversal tools improve the quality, speed and interpretability of Root-Cause-Analysis under dynamic cluster conditions compared to snapshot-based approaches."

Methodology

Delta Queue :

- Buffering stream [2]
- Selection and compaction [6, 4]

Graph Planer [7]:

- Aggregation, ordering and efficient planning of updates

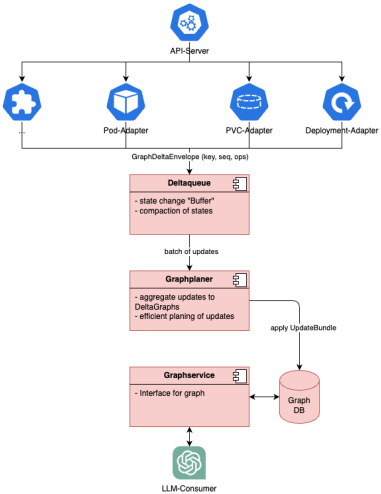

Figure 4: Proposed Architecture

Methodology

Graph service:

- graph mine initial context
- refine context with predefined traversal tool
- trace trajectory for interpretability [1]

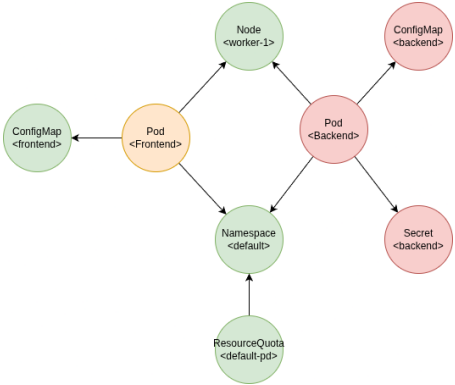

Figure 5: Example graph

Evaluation

Fault injection:

- Dataset of synthetic workflows with malformed manifests
- Each workflow:
  - apply to cluster
  - run RCA
  - cleanup
- Reproducible across both systems for fair comparison
- Example: *Deployment with invalid image tag*

Ablations parameters:

- Graphplaner:
  - batch or interval based
  - batch size and interval time
- Graph traversal:
  - no graph mining / no refinement
  - different graph mining algorithms

Metrics:

- resemble Synergy-RCA metrics
  - time-to-error-detection
  - correctness
  - token-usage

Contributions

- live-state graph approach under dynamic conditions
- Introduction of LLM-friendly and interpretable traversal interface
- Design of RCA-Dataset for Kubernetes
- Derives recommendations for future AI-based RCA-Systems

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